

Y.V.S.S.SHUBAM

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SKILLS

- Programming Languages: Python, C++, JavaScript
- Machine Learning: PyTorch, Scikit-learn, Pandas, NumPy, FAISS, Sentence-Transformers
- Backend: Node.js, Express, FastAPI, REST APIs, JWT
- Databases: MongoDB, MySQL
- Tools and Platforms: Git, Docker, Postman, Linux, Google Cloud Platform

PROJECTS

AI-Powered Farming Advisory System

KrishiDost-AI Farming Assistant PyTorch | YOLOv8 | FAISS | Ollama Mistral | FastAPI | Node.js | MongoDB Link: github.com/yvssshubam/KrishiDost

- Built a full-stack farming assistant with three AI services: a PyTorch CNN (PlantVillage, 50K+ images) for disease detection, a Random Forest for NPK-based crop recommendation, and a FAISS-backed RAG chatbot on a local Ollama Mistral LLM.
- Architected a four-service system (Express auth, FastAPI ML, FastAPI RAG, SPA frontend) with JWT, MongoDB, and English/Hindi/Tamil support.
- Designed a RAG pipeline with Sentence-Transformers embeddings over 500+ agri documents and top-k retrieval to ground responses and cut hallucinations.
- Shipped live mandi prices, government scheme lookup, and a community forum.

Stroke Risk Prediction System

Python | Scikit-learn | Pandas | NumPy | Matplotlib

- Built a classification pipeline on 5,000+ patient records using SMOTE oversampling, one-hot encoding, and 5-fold cross-validation across Logistic Regression, Random Forest, and Gradient Boosting.
- Achieved 92% recall on the minority class with a tuned Random Forest model, prioritizing sensitivity for clinical prediction.
- Generated feature importance analysis and confusion matrix visualizations for model interpretability.

Employee Performance & Bonus Management System

C++ | OOP | File I/O

- Designed a modular system using inheritance, abstract classes, and polymorphism to model employee roles with dynamic bonus calculation based on weighted KPIs.
- Built a file-based persistence layer for employee records and CSV-exportable performance reports.

EDUCATION

Bachelor of Technology (B.Tech) in Computer Science and Engineering

Vellore Institute of Technology (VIT)

CGPA:8.5

Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Machine Learning, Probability and Statistics, Discrete Mathematics

CERTIFICATES

Machine Learning Specialization – Linear Regression — Issued by DeepLearning.AI 🎓 (Coursera, 2026). Covered supervised learning, cost functions, gradient descent optimization, and model evaluation.

Advanced Learning Algorithms — Issued by DeepLearning.AI 🎓 (Coursera, 2026). Covered neural networks, decision trees, ensemble methods, and best practices for ML system development.

Google Cloud Skills Certification — Completed hands-on labs on cloud infrastructure provisioning, App Engine deployment, and GCP networking.

IoT and Digital Transformation — Covered IoT architecture, cloud integration, edge computing, and digital systems transformation.

Network Basics Certification — Studied TCP/IP, routing, switching, and networking fundamentals